Human influences On ecosystem

Human population increased because of:
- Medical revolution
- Agricultural revolution.

① Deforestation: Negative impacts on ecosystem:
1- Less/no trees → Less photosynthesis → less carbon dioxide removed from the atmosphere → CO_2 levels increases in the atmosphere → Causes global warming because CO_2 is a greenhouse gas.
2- Less trees → less transpiration → Less rainfall
3- Less trees → less roots to hold soil particles → leads to soil errosion → leads to floods → Wash out of minerals → Soil infertility.
4- Damage of habitats → animals become extinct → Disturbs food chains → Decrease biodiversity

Biodiversity: The number of different species that live in an area.

② Raising more livestock: Negative impacts on the environment:
- More methane gas is produced → Green house gas → Global warming

* Methane gas is produced:
  - Waste gas of cattle
  - Rice fields
  - Decay of rubbish

* Greenhouse gases:
  - Water vapour
  - Methane
  - Carbon dioxide

→ Cause global warming: An average increase in temperature of earth surface

3- Eutrophication by
→ Untreated sewage
→ Fertilizers.

→ Leakage of untreated sewage and fertilizers into water in rivers will increase the concentration of nitrate ions in water, which will increase the growth of algea. The algea will float and block sunlight, so producers or plants which live at the bottom of the river cannot photosynthesize, and will die. Dead plants will be decomposed by decomposers like bacteria. Bacteria uses dissolved oxygen in water for its respiration. So levels of dissolved oxygen in water will decrease, so other aquatic animals will die as there is no enough oxygen.

4- Waste
→ Biodegradable: Can be decomposed
→ Non-biodegradable: Cannot be decompered. ↓ plastic
  → Release harmful chemicals into soil; plants & microbes die
  → Animals get trapped in plastic or entangled, leading to injury or inability to move
  → Injesting plastics can internally injure, poison, or block the animal's digestive system
  → Harmful micro-plastics are injested and accumulate in the food chain

5- Mono-culture

6- Non-sustainable resources
→ Fish stock
→ Fossil fuels

7- Destruction of habitats: Reasons:
1- Increased area for housing, crop plant production, and livestock production
2- Extraction of natural resources (mining, logging and drilling)
3- Fresh water and marine pollution

How humans increased food production:
1- Agricultural machinary to use larger areas of land and improve efficiency.
2- Chemical fertilizers to improve yields.
3- Insecticides to improve quality and yield.

4- Herbicides to reduce competition with weeds.
5- Selective breeding to improve production by crop plants and livestock.

Sustainable resource: One which is produced rapidly as it is removed from the environment so that it does not run out.

Why organisms become endangered?
1- Climate change
2- Habitat destruction
3- Hunting
4- Overharvesting
5- Pollution
6- Introduced species.

How endangered species can become conserved?
1- Monitoring and protecting species and habitats
2- Education
3- Captive breeding programmes
4- Seed banks.

How forests can be conserved?
1- Education
2- Protected areas
3- Quotas
4- Replanting

How fish stock can be conserved?
1- Education
2- Closed seasons
3- Protected areas
4- Controlled net types and mesh size
5- Quotas
6- Monitoring

What are the reasons for conservation programmes?
1- Maintaining or increasing biodiversity
2- Reducing extinction
3- Protecting vulnerable ecosystems
4- Maintaining ecosystem functions: * Nutrient cycling
* Resource provision: - Food
- Drugs
- Fuel
- Genes

For endangered animals, captive breeding programmes in zoos are made to increase genetic variation.
1- AI: Artificial insemination: Collecting sperm from male animals and introduce it in female animal at time of ovulation
2- IVF: in vitro fertilization: The fertilization of an egg outside of the animal; eggs are extracted and mixed with sperm. When the zygote forms, it's reinserted into the uterus of the female.

Monoculture: The cultivation of a singal crop in a given area.

Advantages of large scale monocultures of crop plants:

1- Higher yield because of: - Less competition with other species
- Conditions are optimised for a single plant

2- More efficient because ploughing, planting and harvesting are the same across a wide area.

3- More profitable because of: - Bigger harvests
- Less labour intensive

Disadvantages:

1- Lack of variation; a disease that infects one individual can afflict the entire population.

2- Loss of nutrients because crops are removed; nutrients would otherwise be put back into the soil by decomposers.

3- Displacement of species as native plants and animals are killed or forced to relocate

4- Soil erosion; soil is washed into rivers and streams, alongside harmful pesticides.

Advantages of intensive livestock productio:

1- Higher yields / profits: - Conditions are optimised for production
- More efficient.

2- Easier to manage: - Less land is required
- Supervision of animals is easier

3- More profitable: - Production is cheaper
- Automation reduces labour

Disadvantages:

1- Soil erosion: Overgazing damages the grass and exposes soil.

2- Disease: - Animals are kept in close proximity
- Animals suffer and diseases spread easily

3- Eutrophication: - Animal waste causes algal growth
- Fish die in deoxygenated water

Sources of CO_2:

- Burning of fossil fuels
- Deforestation
- Respiration in animals.

Explain the risks to species if population size decreases:

Smaller populations carry less genetic variation

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Because fewer organisms mean fewer traits / characteristics in the gene pool

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Affects the specie's ability to cope with environmental changes in the future.

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Putting it at greater risks of extinction